THE UNIVERSITY OF SYDNEY

BMET5934 Lab Report

Feature-Based Physiology for Sleep Apnoea Detection

Faculty of Engineering

Group 17
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Abstract

This report outlines the development of an apnoea screening system based upon the pre-existing ECG and SpO2 measurements during 100 patients' overnight sleep recordings. Supplied QRS indices were utilised to calculate RR interval series, and alongside SpO2 traces, they were both cleaned by converting physiologically unreasonable values to NaN. The first submission used a sliding 30 s window with a 1 s hop, extracting six features (meanRR, stdRR, NN50, SDSD, meanSpO2, stdSpO2) and labelling each window from its end point. A fixed linear regression model with an F1-optimised threshold, chosen on the training data only, achieved an F1 of about 47.7% in cross-validation and 44.3% on the hidden test set. In the second submission, the same six features were recomputed over three consecutive 30 s windows (past, current, future) and concatenated, with labels taken from the end of the future window to better match the timing of apnoea and desaturation. Using the same linear classifier and thresholding procedure, this 3-epoch design improved performance to roughly 49.8% F1 in cross-validation and 47.0% on the hidden set. The report details the data issues encountered, the reasoning behind these design choices, and how the reported evaluation separates genuine improvement from overfitting.

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Database Description

The training dataset comprised 100 patients anonymised overnight recordings. For each patient we are provided with the continuous ECG, a SpO2 trace, expert QRS annotations, and a per-epoch apnoea/normal label sequence.

- ECG: sampled at 200 Hz; Integer counts, units treated as arbitrary amplitude in microvolts
- SpO2: sampled at 1 Hz; Integer counts of a percentage oxygen saturation
- Provided variables per subject:

ECG{p}: ECG samples; SpO2{p}: SpO2 samples; QRS{p}: R-peak sample indices; Class{p}: 1 second resolution apnoea annotations: N: 'normal', A: 'apnoea'.

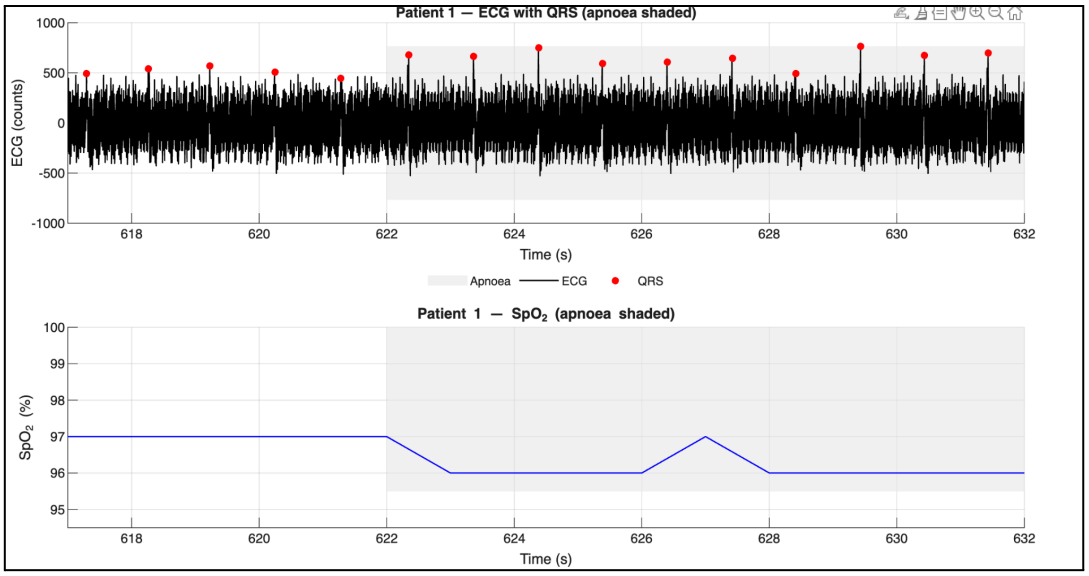

Figure 1: Example 15 s segment from Patient 4 showing ECG with detected QRS complexes and SpO2. Grey band indicates intervals labelled as sleep apnoea.

Table 1: Description of inherent dataset challenges

Issue #	Challenge
1	Mismatched sampling rates: ECG is recorded at 200 Hz while SpO2 and labels are at 1 Hz, making joint feature extraction non-trivial.
2	SpO2 artefacts and unrealistic values: SpO2 traces contain dropouts, and physiologically impossible values, requiring cleaning.
3	Physiological lag in SpO2: desaturation often lags behind apnoea-related ECG changes by several seconds, so using synchronous SpO2 values can misrepresent causal timing.
4	Inconsistent class prevalent across patients: some patients are apnoea-heavy while others contain little or no apnoea, leading to potential bias in per-patient CV.
5	Class imbalance: apnoea epochs form only a small fraction of labelled seconds, resulting in a heavily skewed A/N ratio

Overview Diagram

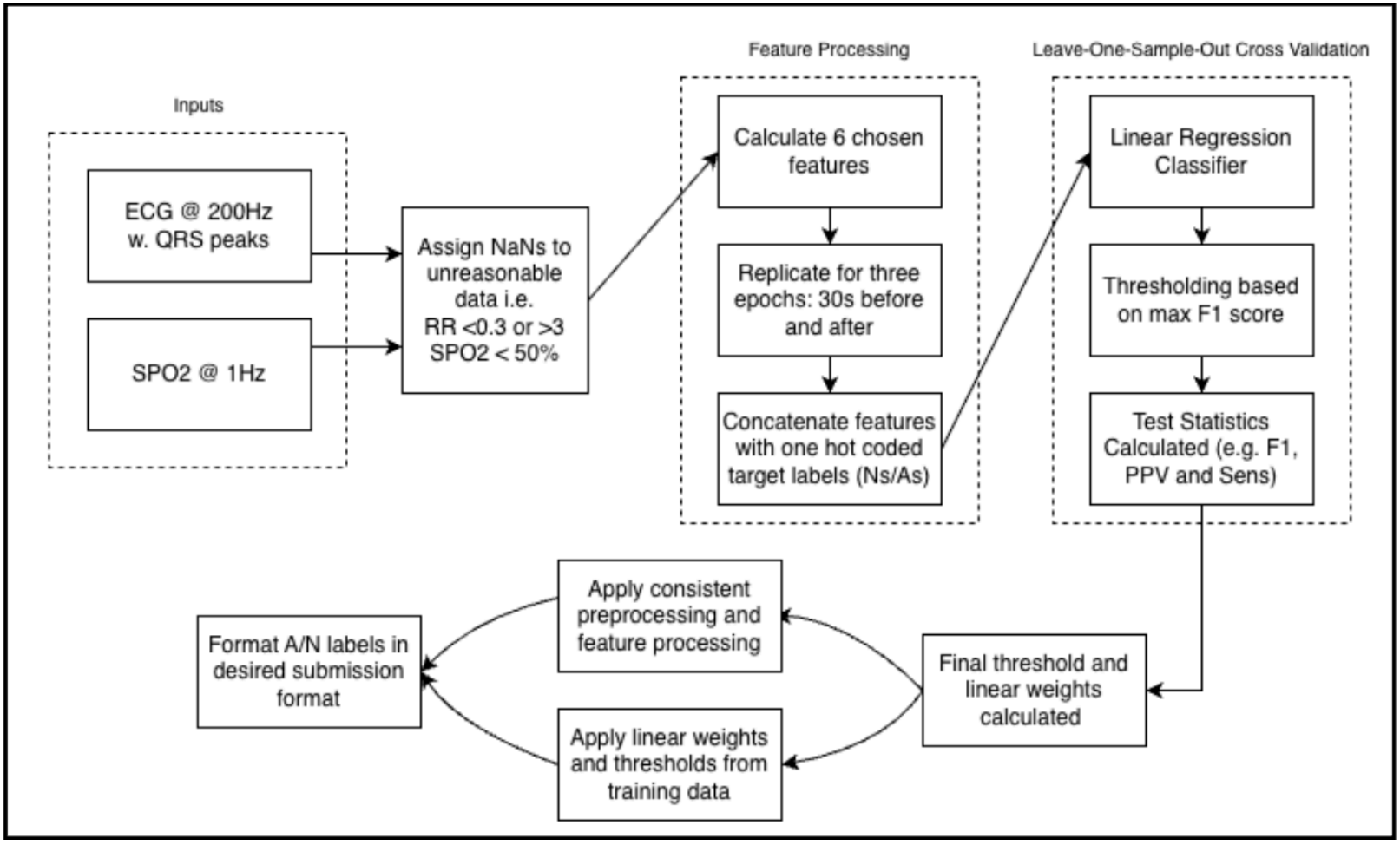

Figure 2: Overview diagram outlining a high level description of the model system for sleep apnoea detection, including preprocessing, feature calculation, leave-one-out cross validation, and thresholding for maximum F1 score before application to the hidden test set.

System Performance

Table 2: Performance results for submissions 1 and 2

	XV on the 100 training set records			Performance on the 100 hidden set records		
Submission	**Sens (%)**	**PPV (%)**	**F1-score (%)**	**Sens (%)**	**PPV (%)**	**F1-score (%)**
1	62.13	38.74	47.72	61.55	34.57	44.28
2	65.05	40.35	49.80	63.85	37.13	46.96

Submission 1:
Submission 1 uses a single 30 s window with 1 s hop, a compact feature set (4 HRV metrics + 2 SpO2 statistics), and a linear regression classifier with an F1-optimised threshold chosen on the training set. It achieves an XV F1 of 47.7% and 44.3% on the hidden set. The modest drop is expected:

1. **Prevalence shift**. The decision threshold is tuned to the apnoea/non-apnoea ratio in the training cohort. If the hidden set has a different apnoea burden, the same threshold no longer sits at the optimal point on the precision–recall curve, typically depressing PPV or sensitivity.
2. **Signal-quality mismatch**. Submission 1 is sensitive to how RR outliers are removed and how SpO2 NaNs are handled. Different QRS error modes or SpO2 artefacts in the hidden set perturb these 6 summary features, creating both extra false positives (noisy HRV/SpO2 mimicking events) and false negatives (epochs dropped or flattened by artefact).
3. **Limited temporal context**. As each prediction sees only a local 30 s slice, events in the hidden set differ from those implicitly baked into the threshold on the training set, making the model more brittle.

Submission 2:
Submission 2 refines the same linear backbone. Apnoea labels are using three consecutive 30 s windows (past, current, future). Thresholds are selected via patient-wise CV with an F1 sweep.

This yields higher XV performance (F1 ≈ 49.8%) and a corresponding improvement on the hidden set (F1 ≈ 47.0%) due to:

- The 3-epoch design matches the physiological sequence, improving causal alignment.
- RR and SpO2 being cleaned without destroying timing information, making the model less sensitive to dataset-specific artefacts.
- The richer but still linear feature space separating apnoeic vs normal structure without relying on a high-capacity classifier.

A residual XV-to-test gap remains for Submission 2 for similar reasons as Submission 1, but now more subtly: the F1-optimal threshold is still chosen on the training prevalence and artefact mix; any shift in apnoea burden, noise patterns or SpO2 behaviour in the hidden cohort moves the true operating point away from that optimum. In addition, the expanded feature set can capture nuances of the training cohort that are not perfectly replicated in the hidden set, leading to small calibration errors even though overall generalisation has improved.

Introduction

I implemented the full signal-processing and classification model used in our final submission. Starting from the raw ProjectTrainData cell arrays, I wrote the code to load and validate the ECG, QRS, SpO_2 and Class signals, and applied simple cleaning while preserving timing. SpO2 values outside a plausible physiological range were set to NaN rather than removed, and RR intervals were derived directly from the supplied QRS indices, with implausible intervals (shorter than 0.3 s or longer than 3.0 s) flagged as NaN so suspicious beats were marked without shifting the time axis.

For feature extraction, I designed and coded the sliding 30 s window with a 1 s hop. In Submission 1, each window produced six interpretable features: meanRR, stdRR, NN50, SDSD, meanSpO2 and stdSpO2, labelled using the state at the end of the window. In Submission 2, I extended this to a three-epoch structure around each decision point, concatenating the same six features from the past, current and future 30 s segments to better capture how apnoea unfolds over time.

For classification, I used the provided linear regression model (TrainLR) with a bias term, applied via PredLin to obtain consistent apnoea and normal scores. I implemented the patient-wise leave-one-out-cross-validation, the choose_threshold_on_train routine to select an F1-maximising threshold from training scores only, and the final script that retrained on all 100 records and produced the metrics submitted for evaluation.

Development Description

Sensor signal preprocessing

For preprocessing I kept things simple and explicit. For SpO2, I applied one rule for everyone: any value below 50% or above 100% was treated as invalid and set to NaN. For RR intervals from the ECG, I kept the original QRS timestamps but marked intervals shorter than 0.3 s or longer than 3.0 s as NaN so they were excluded from HRV-style features without shifting time. This keeps all channels aligned to the same timeline as the labels and removes points that are clearly non-physiological rather than quietly reshaping the signal.

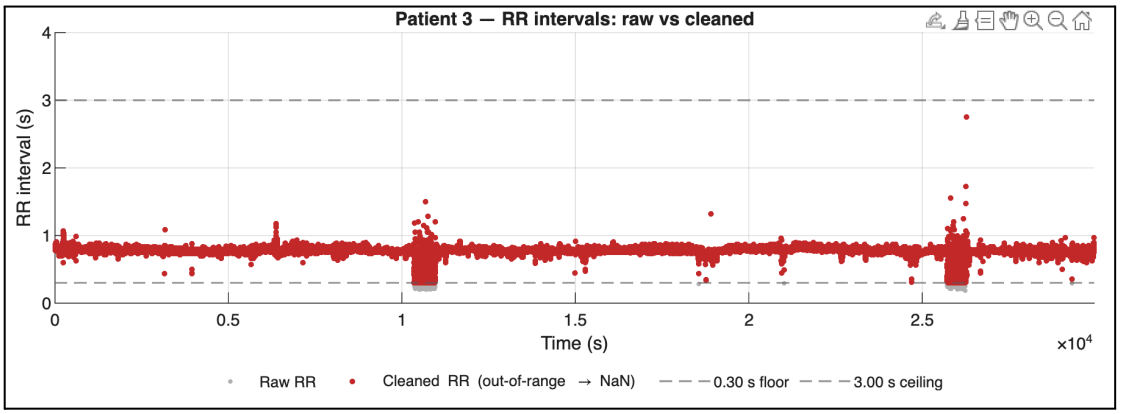

Figure 3: Visualisation of RR interval boundaries for patient 3 instance, ensuring physiologically implausible values are removed.

To check that this behaved sensibly, I visualised raw vs cleaned RR intervals for a representative patient (patient 3). The figure shows that the cleaning mainly knocks out isolated spikes and very long gaps, while the main band of RR values is preserved. That gave me confidence I was cutting obvious detector errors instead of real beat-to-beat variability.

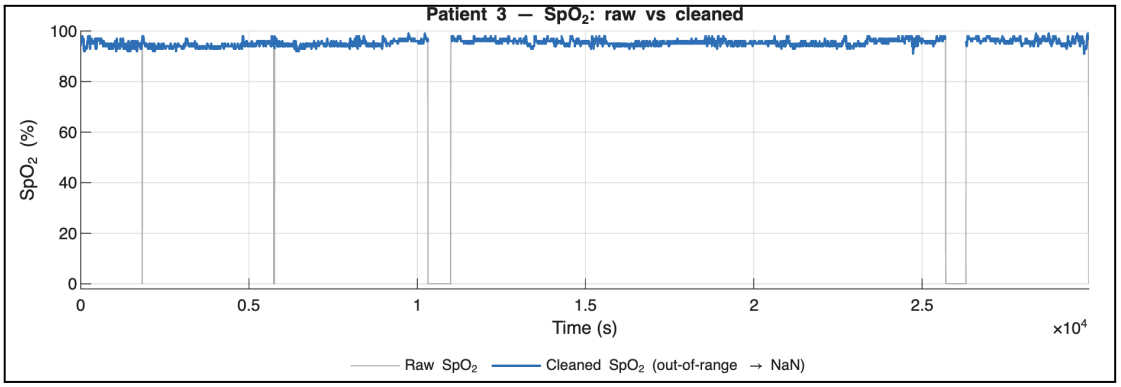

Figure 4: Visualisation of SpO2 cleaning data for patient 3, ensuring physiologically implausible values are removed.

I did the same for SpO2. Plotting the raw and cleaned traces together shows that only extreme outliers are blanked to NaN and that the general shape of the trace, including any desaturation patterns, is unchanged. This confirmed that the rule is conservative enough not to manufacture fake drops or recoveries.

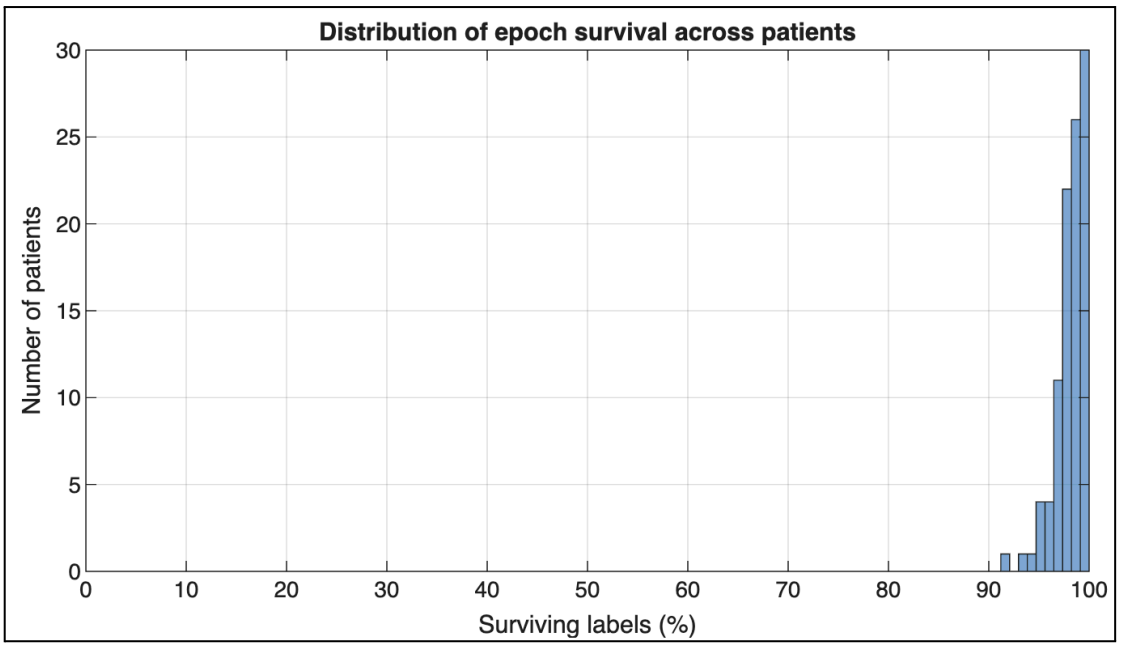

Figure 5: Histogram representation of the percentage of surviving data labels per patient.

After applying RR and SpO2 cleaning and dropping any windows with missing or invalid features, I computed, for each patient, the percentage of candidate 1 s epochs that remained usable. Figure 5 shows their distribution. Most nights retain well over 90% of potential epochs, confirming that the cleaning rules remove bad data without collapsing coverage for individual patients.

QRS/RR construction

For QRS/RR construction I treated the supplied QRS indices as the timing reference and built beat-to-beat features from there. For each patient I converted consecutive QRS locations into RR intervals in seconds and, in parallel, computed midpoint timestamps in ECG samples for each interval. I then applied a simple plausibility rule: RR values shorter than 0.3 s or longer than 3.0 s were marked as NaN but not removed. This keeps the original time axis intact for later 30 s windows and 1 Hz labels, while clearly flagging intervals that are physiologically implausible or likely due to missed or extra QRS detections.

```
%% === RR build (seconds) + RR midpoints (samples) ===
RR = cell(F,1); RRmid = cell(F,1);
for p = 1:F
   q = QRS{p}(:);
   if numel(q) < 2
       RR{p} = []; RRmid{p} = [];
       continue
   end
   rr  = diff(q)/Fs;                    % seconds
   bad = (rr < 0.3) | (rr > 3.0);       % out-of-range
   rr(bad) = NaN;                       % mark as NaN
   qm = (q(1:end-1) + q(2:end))/2;      % samples (midpoint timestamp)
   RR{p}   = rr;                        % keep NaNs (no filtering)
   RRmid{p}= qm;                        % keep all midpoints (no
subsetting)
end
```

These choices were made to keep the RR series usable and honest at the same time. Using midpoints gives each RR interval a well-defined location in time, so when I slide windows later, RR-based features line up cleanly with both the ECG trace and the Class labels. Keeping implausible intervals as NaNs (rather than deleting points) avoids time warping: windows that overlap bad beats simply see missing RR information instead of being silently shifted, which is safer for anything that depends on temporal context.

To check that this behaved as intended, I spot-inspected several patients visually. For each, I plotted a short ECG segment with QRS markers overlaid, and underneath plotted the corresponding RR time series derived from those markers using the rules above. In normal rhythm the markers sit neatly on R peaks and the RR trace stays within the physiological band; when the detector hiccups or the signal is noisy, the affected intervals appear as spikes beyond the 0.3 to 3.0 s limits and are correctly turned into NaNs in the RR plot. This confirmed that (i) QRS placement was reasonable in the regions used for features, and (ii) the plausibility gating was catching obvious errors without distorting the timing of valid beats.

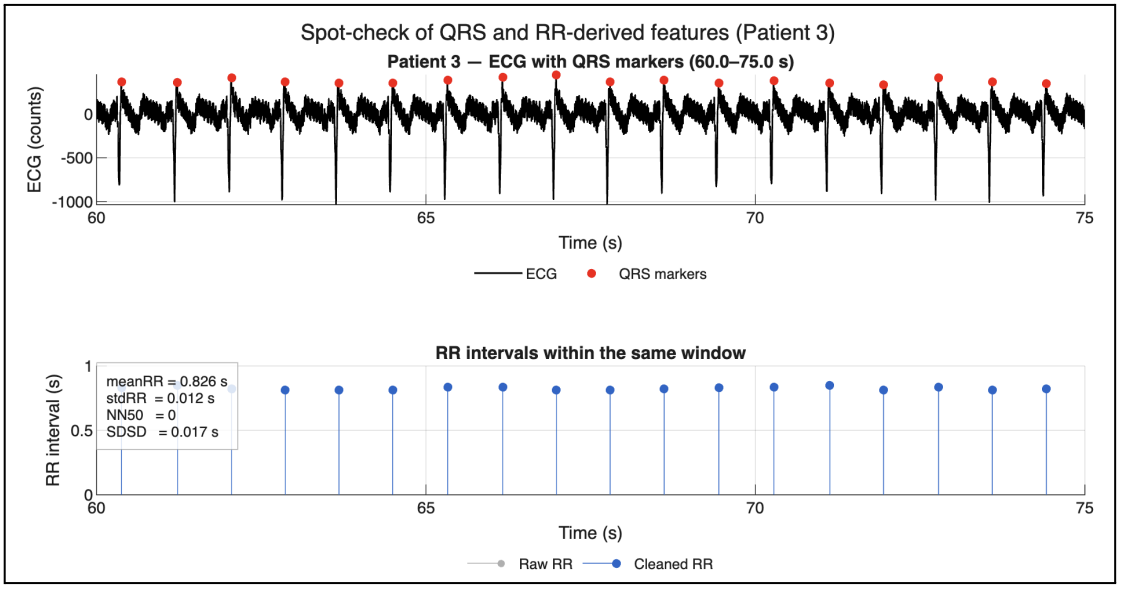

Figure 6: Spot-check of QRS alignment and RR feature calculation for patient 3.

Feature extraction

3.1 Submission 1

For the first submission I built a straightforward local descriptor at 1 s resolution. I slid a 30 s window through each record with a 1 s hop. Within each window I used the cleaned RR series and SpO2 to compute six scalar features: meanRR, stdRR, NN50, SDSD (beat-to-beat variability), and meanSpO2 and stdSpO2. Each 30 s window was given the label taken from the end of that window, so every feature vector summarised “the last 30 seconds” leading up to the annotated state (A or N). This produced a dense set of labelled 30 s epochs across all patients using only simple, interpretable statistics.

The reasoning was to start with something transparent and quick: short-term HRV and basic SpO2 variability are standard markers of respiratory disturbance, and using a single 30 s slice keeps the mapping between signals and labels easy to understand. It also created a clean baseline for later changes: any improvement from more context or different framing could be compared directly back to this simple six-feature setup.

To test this baseline, I trained the provided linear regression classifier on all 100 training records using these features, chose an F1-optimal threshold on the training scores, and recorded the resulting F1, sensitivity and PPV as my initial reference performance. Finally, I submitted this model to the hidden test set; the drop from cross-validation to hidden performance was used later to judge whether subsequent design changes (the 3-epoch features in Submission 2) were genuinely improving generalisation rather than just fitting quirks of this first setup. The performance comparison between the training set and hidden test set as seen in Figure 7 below displays that the current use of features is holding strong performance on unseen datasets, justifying their choice in this submission.

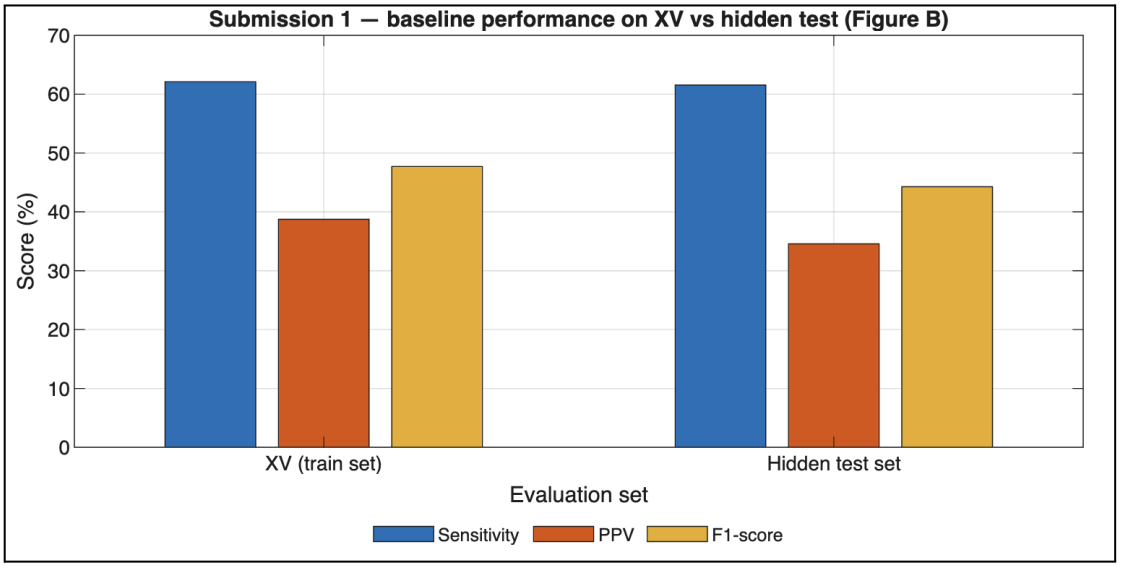

Figure 7: Submission 1 baseline performance on cross-validation vs hidden test set (Sensitivity, PPV and F1-score).

3.2 Submission 2

For the second submission I kept the same simple linear classifier but changed how each decision point is represented. Around every time (t) I built three consecutive 30 s windows: a past context window (-30 to 0 s), the central window (0 to 30 s), and a future window (30 to 60 s). For each window I recomputed the same six features as Submission 1 (meanRR, stdRR, NN50, SDSD, meanSpO2, stdSpO2), then concatenated them to form an 18-dimensional feature vector. The target label for this vector is taken from the end of the future window (the end of 30-60 s), so each feature set describes how RR and SpO2 behave around the point where apnoea or normal breathing is declared. Figure 8 from our Week 11 update highlights the epoch structure which inspired the creation of this submission structure.

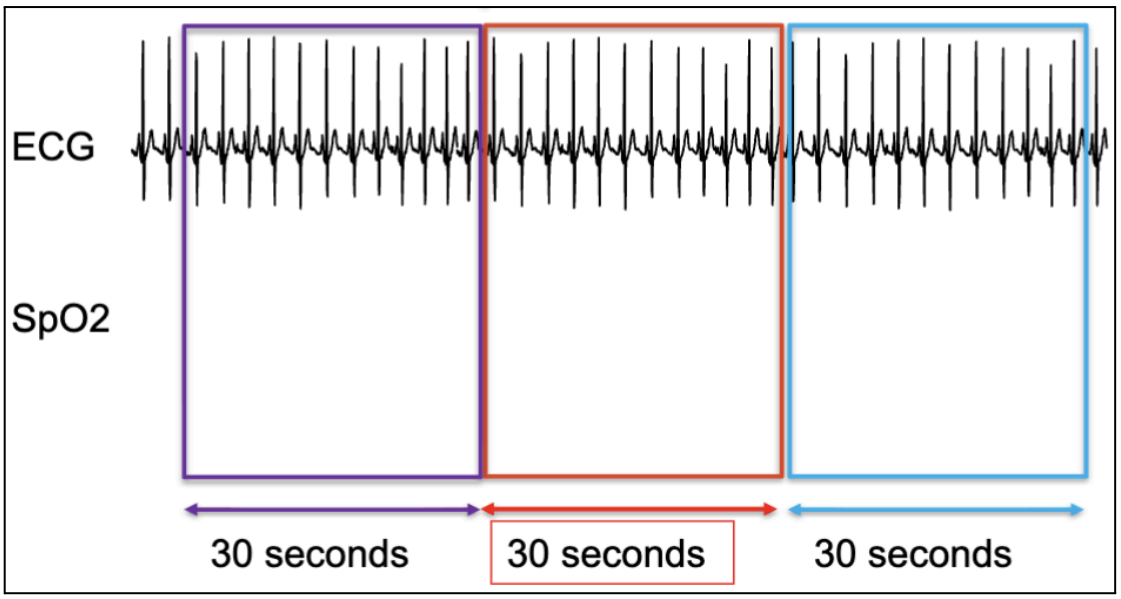

Figure 8: Three-epoch layout, following the schematic provided in the BMET5934 project slides (BMET5934_MajoProject_2025_LabWeek11update.pptx).

This design lines up better with the physiology of obstructive events. Apnoeas are not isolated points: RR variability often changes before and during the event, while SpO2 tends to drop with a delay. By tying the label at 30-60 s to features drawn from -30 to 60 s, the model can "see" the build-up, the event, and the lagged desaturation pattern without changing classifier complexity. It is still a linear model over hand-crafted features, but each decision now reflects a short narrative in the signals rather than a single snapshot. The implementation mirrors this idea directly in code by constructing three adjacent windows per anchor point, computing the six features in each, concatenating them [fPast, fCurr, fFutur], and using segF_end (the end of the future window) to index the Class label, as shown below:

```
for j = 1:Nwin
        seg0_start = (j-1)*Step + 1;                % 0:30 start
        seg0_end   = seg0_start + WindowLen - 1;    % 0:30 end
        segP_start = seg0_start - WindowLen;        % -30:0
        segP_end   = seg0_start - 1;
        segF_start = seg0_end + 1;                  % 30:60
        segF_end   = seg0_end + WindowLen;
        fPast   = feats(segP_start, segP_end);
        fCurr   = feats(seg0_start, seg0_end);
        fFutur  = feats(segF_start, segF_end);
        Xi(j,:) = [fPast, fCurr, fFutur];
        % Label from END of 30:60 epoch (future framing)
        lblIdx = floor(segF_end / Fs);
        if lblIdx <= numel(Class{p})
            c = Class{p}(lblIdx);
            if     c=='A', Ti(j,:) = [1 0];
            elseif c=='N', Ti(j,:) = [0 1];
            end
        end
    end
```

To evaluate this change I reapplied the same patient-level cross-validation procedure used for Submission 1. For every fold I rebuilt the 3-epoch features on the training patients only, trained the linear regression classifier, and chose the decision threshold on the training scores via an F1 sweep. I then froze that threshold and measured sensitivity, PPV and F1 on the held-out patients. I also trained once on all 100 training records to obtain full-set metrics, and

submitted the corresponding model to the hidden test server. As summarised in Figure 9, the 3-epoch representation improved both cross-validated and hidden-set F1 (from about 47.7% to 49.8% on XV, and from 44.3% to 47.0% on the hidden cohort), showing that adding structured temporal context with the same six features produces a genuine and stable performance gain rather than simply fitting quirks of the training data.

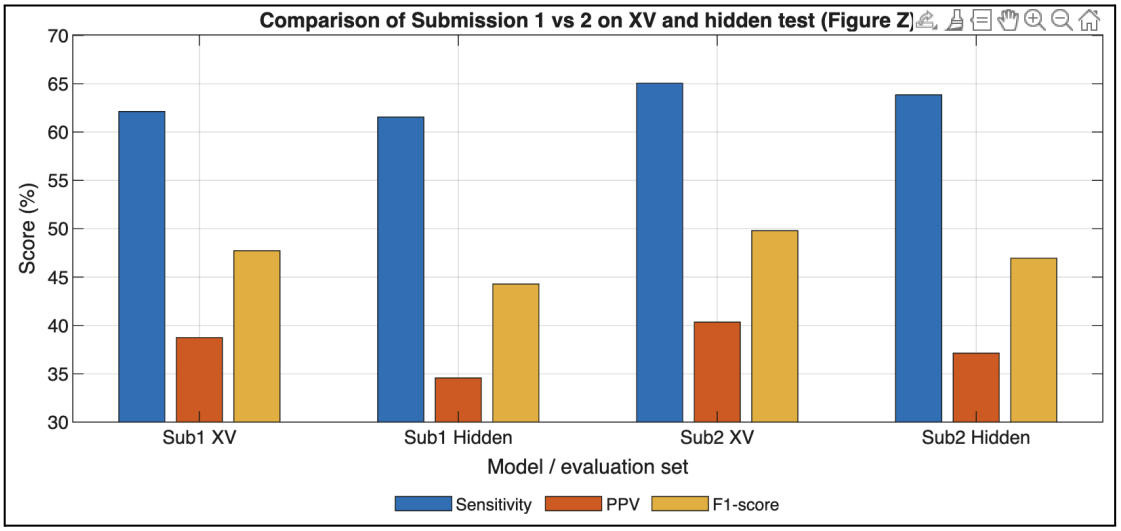

Figure 9: Comparison of Submission 1 and Submission 2 performance (Sensitivity, PPV and F1-score) on cross-validation (XV) and hidden test sets.

Classifier & output thresholding

For both submissions I kept the classifier deliberately basic so that any improvement had to come from how I built the features and aligned them with the labels, not from swapping to something fancy. The model is a linear regression classifier with a bias term (TrainLR). I take the feature matrix X, add a column of ones, and solve the least squares system of Equation 1:

Equation 1:

$$W = (X'X)\backslash(X'T)$$

where T is the two-column one-hot target [A N]. The result is a weight matrix W that turns each feature vector into two scores: one for apnoea and one for normal. In the report I show this directly, both as that closed-form expression and as a short code snippet from TrainLR, to make it clear that training is fully transparent and deterministic.

```
function W=TrainLR(X,T)
%Train a linear regression classifier using data (X) and targets (T)
K=size(X,1);
X=[X,ones(K,1)];        % add bias
W=(X'*X)\(X'*T);        % closed-form LS
end
```

Predictions use a thin wrapper, PredLin. It re-attaches the bias column, multiplies by W to get Y, and, if needed, converts each row into a hard one-hot label by taking the larger of the two scores. For this project the important part is that I also keep the continuous apnoea score, taken as the first column of Y, and use that as the basis for thresholding. A short PredLin snippet in the text shows this flow: compute Y, then either pick argmax for a crisp label or carry Y(:,1) forward as the apnoea confidence. Using the same scoring function everywhere (cross-validation, full training run, and final submission) avoids silent changes in behaviour between experiments.

```
function [T,Y]=PredLin(W,X)
%Predict with linear classifier
K=size(X,1); N=size(W,2);
X=[X,ones(K,1)];        % add bias
Y=X*W;                  % scores
[~,temp]=max(Y,[],2);
T=zeros(K,N);
for i=1:K, T(i,temp(i))=1; end
end
```

The operating point is chosen purely from training data using the helper choose_threshold_on_train. For each fold I collect the apnoea scores on the training patients and their true labels, sweep over candidate cut-points, and compute TP, FP and FN to derive the F1 score at each threshold. The function then selects the threshold that maximises F1 on that fold's training data, with explicit guards for edge cases like "all N" or "all A" so it never collapses or explodes when prevalence is extreme. The same pattern is used once on the full 100-record training set when building the final submitted models. Figure 10 shows the precision–recall curve built from the training-set scores of the final 3-epoch linear model. The black marker is the threshold that maximises F1 on the training data, and the dashed vertical line marks the corresponding recall at that operating point.

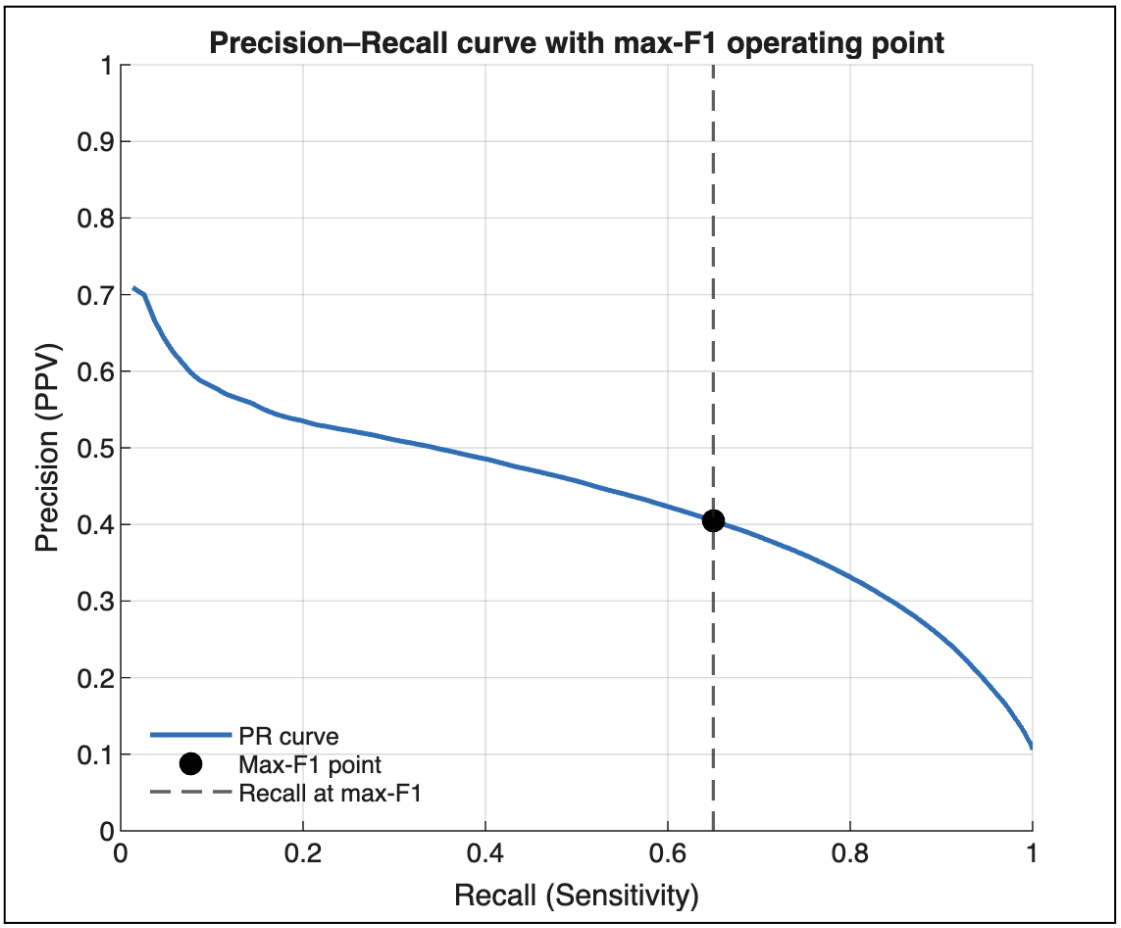

Figure 10: Precision–recall curve for the final 3-epoch linear model, with the chosen operating point (maximum F1) marked and its corresponding recall shown by the vertical dashed line.

Putting this together, every result in Submissions 1 and 2 follows the same recipe: build features, fit W with TrainLR on training patients only, generate scores with PredLin, pick a

single F1-optimal threshold using choose_threshold_on_train on those training scores, then freeze both W and that threshold when evaluating the held-out cross-validation folds and the hidden test set. Because the classifier and thresholding machinery are identical between submissions, the differences in Figure 9 (Submission 1 vs Submission 2 metrics) can be attributed directly to the changed feature framing and use of temporal context, not to a change in model capacity or tuning tricks.

Explored Ideas

First, I experimented with a weighted LDA classifier (TrainLDA) in place of the supplied linear regression. The aim was to handle the strong class imbalance more explicitly by up-weighting apnoea epochs and enforcing a shared covariance model. In code this involved standardising features, applying per-sample weights, estimating class means and a common covariance, and folding priors into the bias term. On paper this is well matched to a linear boundary, but in practice it clashed with the way my model handled NaNs, and with the patient-wise cross-validation. Global z-score scaling had to be recomputed carefully inside each fold to avoid leaking test statistics, and the presence of many NaN entries after RR/SpO2 cleaning made the shared covariance matrix unstable. The resulting decision scores were noisy and hard to calibrate, and did not give a reliable improvement over the simpler combination of TrainLR with a threshold approach, so I dropped LDA from the final system.

I also tested a weighted logistic regression model to see if a probabilistic output with class weights would improve calibration and F1. This required a separate optimisation step and careful feature normalisation. With the available features and heavy imbalance, the fitted models either stayed close to the linear-regression boundary or became sensitive to how badly contaminated windows were filtered. Again, there was no clear or stable gain over the closed-form linear baseline, so I kept the simpler solution.

In parallel, I explored a larger feature set within each 30 s window, including pNN50, RMSSD, median and IQR of RR, log-transformed variability measures, minimum SpO2 and percentage of time below 90%. These features behaved sensibly in “clean” segments but were fragile when there were only a few valid beats or when SpO2 coverage was patchy. Many candidate windows ended up with missing values, and aggressive cleaning to keep them all valid reduced usable training data without a consistent lift in cross-validated performance. For the final submissions I therefore reverted to the smaller, well-behaved six-feature set and focused on the three-epoch framing, which gave a clearer and more reliable improvement.

Future Improvements

Future improvements focus on extending the feature set and exploring slightly richer decision rules while keeping the system interpretable. First, only a small number of simple RR and SpO2 statistics were used; a natural next step is to incorporate more informative descriptors of apnoea-related behaviour. This includes time-domain measures that capture desaturation depth, duration and recovery, counts of repeated desaturations within short windows, and frequency-domain measures of heart rate variability in low and high frequency bands to reflect autonomic changes during disturbed breathing. Second, the work so far used a single linear regression model with a fixed global threshold. Additional classifiers can be tested on the same hand-crafted features, such as logistic regression, regularised linear models, or other shallow models that remain easy to inspect but can provide better calibrated scores. Third, alternative data handling strategies should be evaluated: using simple window-level quality indices, excluding or down-weighting segments with extensive missing or implausible RR or SpO2 values, and checking how these choices affect robustness across patients. Finally, a two-stage approach is worth investigating, in which the first classifier outputs per-epoch apnoea likelihoods and a second, simple post-processing stage enforces temporal consistency (for example, joining very short gaps and removing isolated single-epoch detections) so that predicted events better match realistic apnoea episodes.

AI-Usage

OpenAI's GPT-5 Thinking model was used to help refine the structure, wording and formatting of this report. All methods, code, analyses and conclusions are the author's own work.